

QINZHENG XU

qxu5@tulane.edu • 763-313-6156
Charlotte, NC 28202

EDUCATION

Ph.D. in Economics , Tulane University	2022–2025
<i>Dissertation: Navigating Crisis—The Interplay of Hate Crimes, COVID-19, and Climate Change in Financial, Healthcare, and Energy Systems</i>	
Committee: Mary K. Olson (Chair), Felipe Gómez Trejos, Kevin Callison, Nicholas Mattei	
Master of Finance , Tulane University	2019–2021
Master of Financial Mathematics , University of Minnesota	2017–2019
Bachelor of Economics , Shandong University, China	2013–2017

FIELDS OF INTEREST

Environmental & Energy Economics; Health Economics; Financial Economics

JOB MARKET PAPER

Soft Resilience in Hospitals Under Stress: Evidence from Healthcare-Associated Infections During COVID-19

Abstract: This paper studies when communication-related soft resilience becomes more consequential for infection control performance. I use pre-pandemic hospital communication quality as a proxy for this capacity. The analysis combines HCAHPS, CMS Hospital Compare, and Hospital Cost Report data for 3,366 U.S. hospitals from 2018 to 2021. I use the onset of COVID-19 as a common external stress shock. The empirical design examines whether infection outcomes changed differently after COVID-19 across hospitals with different pre-pandemic communication quality. The main relationship appears in BSI outcomes with higher COVID-induced diagnostic noise. A one-point increase in pre-pandemic communication quality is associated with about a 10 percent smaller post-COVID change in these infection rates relative to their sample means. No comparable pattern appears for non-BSI outcomes with lower COVID-induced diagnostic noise. I then examine two additional margins of effective task-related noise. The relationship is stronger among hospitals with less prior BSI-related experience. It also becomes stronger as local COVID-19 intensity rises for BSI outcomes with higher COVID-induced diagnostic noise. The results are robust to hospital fixed effects, pre-pandemic medical delivery quality, staffing, alternative communication measures, and sample restrictions. Together, the findings show that communication-related soft resilience is more strongly associated with infection-control performance when effective task-related noise is greater.

PUBLICATIONS

- Gomez-Trejos, F., **Xu, Q.**, & Lopez, J. (2026). Do severe storms affect air quality in high-population urban centers? Evidence from extreme weather events in Texas. *Energy Economics*, 109191.
- Gao, F., Qiao, Y., **Xu, Q.**, & Zhang, T. (2026). Racial Animus, Employment Risk, and Analyst Forecasts During the COVID-19 Pandemic. *Financial Review*.
- Lu, Y., Qi, L., **Xu, Q.**, Li, Z., Duan, H., He, F., ... & Hyman, J. M. (2025). ELISA protein detector (EPD): A Python-based ELISA tool for accurate low-level protein quantification. *Journal of Immunological Methods*, 539, 113847.

WORKING PAPERS

Soft Resilience in Hospitals Under Stress: Evidence from Healthcare-Associated Infections During COVID-19

Job Market Paper

Xu, Q.

Do Reliability Markets Alter the Spatial Distribution of Urban Air Pollution?

Gomez-Trejos, F., & **Xu, Q.**

RESEARCH EXPERIENCE

Research Assistant, Tulane University 2019–2025

- Supported empirical research in health economics, financial economics, and energy economics using large administrative, financial, and market datasets.
- Conducted econometric analysis, data construction, web scraping, and computational research using Stata, Python, SQL, SAS, and R.

INDUSTRY EXPERIENCE

Quantitative Financial Analyst, Bank of America, Charlotte, NC 2025–2026

- Developed and maintained macroeconomic scenario models supporting stress testing, capital planning, and enterprise-wide global risk analysis.
- Designed NGFS-based climate and energy-transition models linking transition pathways to macroeconomic and financial risk factors, including GDP, GVA, equity, and credit spreads.
- Enhanced foreign-exchange and interest-rate models through scenario-based sensitivity analysis across alternative macroeconomic conditions.

CONFERENCE PRESENTATIONS & SEMINARS

Association of Environmental and Resource Economists Summer Conference, Santa Ana Pueblo, NM May 2025

American Finance Association Annual Meeting, Poster Session, San Antonio, TX January 2024

APPRM Conference, Atlanta, GA November 2023

Microsoft Research Lunch Talk, Boston, MA September 2023

AWARDS

American Finance Association Travel Grant 2024

TECHNICAL SKILLS

Programming: Stata, Python, R, SQL, SAS, MATLAB, Git.

Econometric & Quantitative Methods: Causal inference, panel data econometrics, time-series analysis, duration models, machine learning, natural language processing, deep learning.

REFERENCES

Mary K. Olson (Chair)

Associate Professor
Department of Economics
Tulane University
molson3@tulane.edu

Felipe Gómez Trejos

Assistant Professor
Department of Economics
Tulane University
fgomeztrejos@tulane.edu

Kevin Callison

Associate Professor
Health Policy and Management
Tulane University
kcallison@tulane.edu

Nicholas Mattei

Associate Professor
School of Science & Engineering
Tulane University
nsmattei@tulane.edu